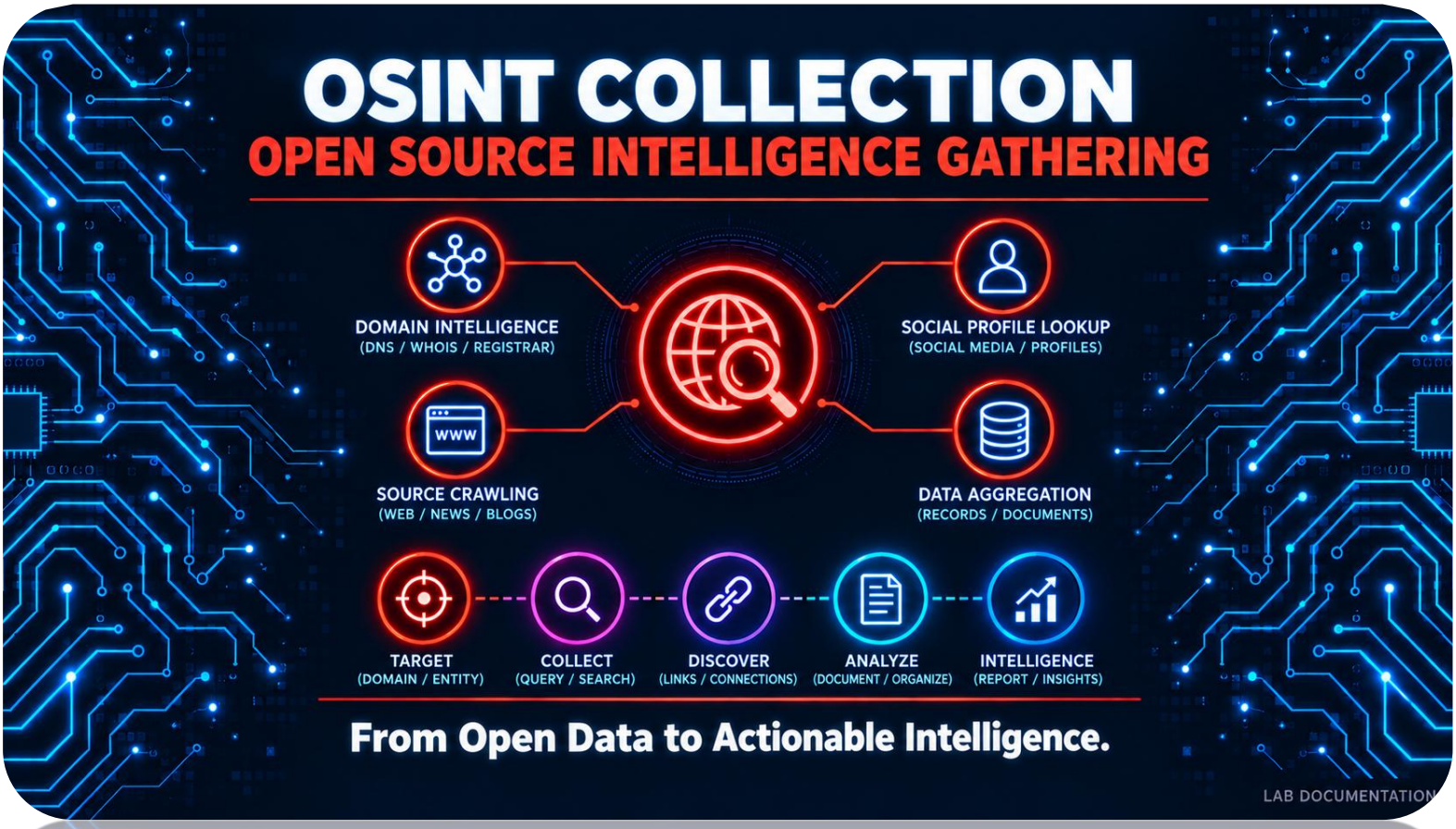


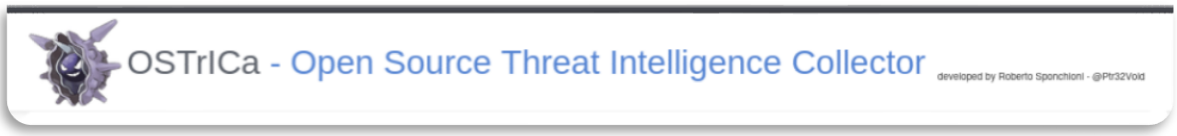
Abdul Wahid

Lab Report

OSINT Collection – OSTrICa & OSRFramework

Open Source Intelligence Gathering Using OSTrICa

OSTrICa (Open Source Threat Intelligence Collector) is a Python-based OSINT tool that collects domain, IP, email, and hash intelligence from multiple online sources and visualizes the relationships in an interactive graph.





Step 1: Install Git

The following command was used to install Git:

```
sudo apt install git
```

Output confirmed Git was already at the latest version (1:2.51.0-1) with no upgrades required.

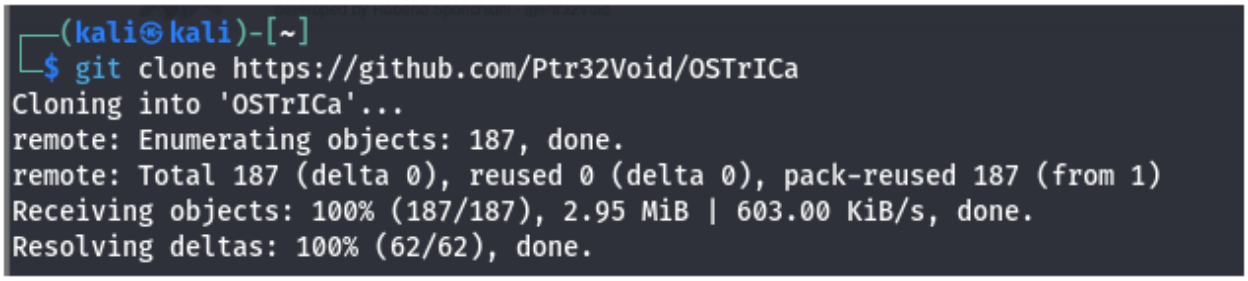


Step 2: Clone OSTrICa from GitHub

OSTrICa was cloned using the following command:

```
git clone https://github.com/Ptr32Void/OSTrICa
```

Git successfully cloned the repository (187 objects, 2.95 MiB). Running `ls` confirmed the OSTrICa folder was created.



Step 3: Navigate to OSTrICa Folder and Execute Tool

Navigated into the folder and launched OSTrICa using Python 2 (required due to Python 2 syntax in the tool):

```
cd OSTrICa
python2 main.py
```

OSTrICa v0.5 loaded successfully, displaying:

- Available plugins: SafeBrowsing, ThreatCrowd, PyWhois, DomainBigData, BlackLists, ThreatMiner, NortonSafeWeb
- Note: Plugins DeepViz, CymruWhois failed to load due to missing modules (deepviz.sandbox, whois, dns)



```
┌──(kali㉿kali)-[~/OSTrICa]
└─$ python2 main.py
OSTrICa v.0.5 - Open Source Threat Intellicence Collector
Developed by: Roberto Sponchioni - @Ptr32Void <rsponchioni@yahoo.it>
write "help" for help
Plugin location: /home/kali/OSTrICa/ostrica/Plugins/DeepViz (DeepViz)
No module named deepviz.sandbox
Plugin location: /home/kali/OSTrICa/ostrica/Plugins/SafeBrowsing (SafeBrowsing)
Loading SafeBrowsing
Plugin location: /home/kali/OSTrICa/ostrica/Plugins/ThreatCrowd (ThreatCrowd)
Loading ThreatCrowd
Plugin location: /home/kali/OSTrICa/ostrica/Plugins/PyWhois (PyWhois)
No module named whois
Plugin location: /home/kali/OSTrICa/ostrica/Plugins/CymruWhois (CymruWhois)
No module named dns
Plugin location: /home/kali/OSTrICa/ostrica/Plugins/DomainBigData (DomainBigData)
Loading DomainBigData
Plugin location: /home/kali/OSTrICa/ostrica/Plugins/BlackLists (BlackLists)
Loading BlackLists
Plugin location: /home/kali/OSTrICa/ostrica/Plugins/ThreatMiner (ThreatMiner)
Loading ThreatMiner
Plugin location: /home/kali/OSTrICa/ostrica/Plugins/NortonSafeWeb (NortonSafeWeb)
Loading NortonSafeWeb
```

Step 4: Explore Available Options Using help Command

Typed help to view all available commands.

```
> help
Following options are available

        domain - used to collect domains information
        Example: domain=google.com or domain=google.com,yahoo.com
        ip - used to collect IP information
        Example: ip=8.8.8.8 or ip=8.8.8.8,173.194.68.99
        md5 - used to collect MD5 information
        sha256 - used to collect SHA256 information
        asn - used to collect ASN information
        email - used to collect email information
        graph - generate a graph based on all the information collected
        cola_graph - generate a graph based on all the information collected where nodes do not overlap (it might tak
e a while to generate the graph if there are lots of nodes)
        gclean - clear graph information
        show - show all information that will be collected
        run - extract intelligece information
        help - this help
        plugins - show available plugins
```

Step 5: Extract Intelligence for www.certifiedhacker.com

The following commands were entered to set the target domain and run intelligence extraction:

```
> domain=www.certifiedhacker.com
> run
```

OSTrICa queried all loaded plugins against the domain. SafeBrowsing and ThreatMiner ran successfully. ThreatCrowd encountered an SSL certificate error (hostname mismatch for otxb.io), which caused it to skip.

```
> domain=www.certifiedhacker.com
> run
Working on www.certifiedhacker.com
Requested intelligence method domain_information is valid for SafeBrowsing!
Running SafeBrowsing() on www.certifiedhacker.com
cleanup SafeBrowsing...
Requested intelligence method domain_information is valid for ThreatCrowd!
Running ThreatCrowd() on www.certifiedhacker.com
ThreatCrowd limit is set (as per limits described on GitHub ThreatCrowd ApiV2)
call_plugin_method() exception HTTPSConnectionPool(host='www.threatcrowd.org', port=443): Max retries exceeded with u
rl: /searchApi/v2/domain/report/?domain=www.certifiedhacker.com (Caused by SSLError(CertificateError("hostname 'www.t
hreatcrowd.org' doesn't match '*.otxb.io'",),)) (ThreatCrowd)!
Traceback (most recent call last):
  File "/home/kali/OSTrICa/ostrica/utilities/Ostrica.py", line 98, in call_plugin_method
    self.save_intelligence(intelligence_information, intelligence_type, plugin['name'], plugin['plugin'].run(intellig
ence_information, extraction_type))
  File "/home/kali/OSTrICa/ostrica/Plugins/ThreatCrowd/__init__.py", line 99, in run
    if intel_collector.extract_intelligence('domain', intelligence) is not None:
  File "/home/kali/OSTrICa/ostrica/Plugins/ThreatCrowd/__init__.py", line 77, in extract_intelligence
    returned_intel = json.loads(requests.get(query, params={typology: intel}).text)
  File "/usr/share/offsec-awae-wheels/requests-2.23.0-py2.py3-none-any.whl/requests/api.py", line 76, in get
    return request('get', url, params=params, **kwargs)
  File "/usr/share/offsec-awae-wheels/requests-2.23.0-py2.py3-none-any.whl/requests/api.py", line 61, in request
    return session.request(method=method, url=url, **kwargs)
  File "/usr/share/offsec-awae-wheels/requests-2.23.0-py2.py3-none-any.whl/requests/sessions.py", line 530, in reques
t
    resp = self.send(prep, **send_kwargs)
```

Step 6: Generate Intelligence Graph

After extraction completed, the graph command was used to generate a visualization:

```
> graph
```

```
> graph
Graph generated in /home/kali/OSTrICa/viz/21840779-75ee-41d4-907d-e249b6f335bf.html
```

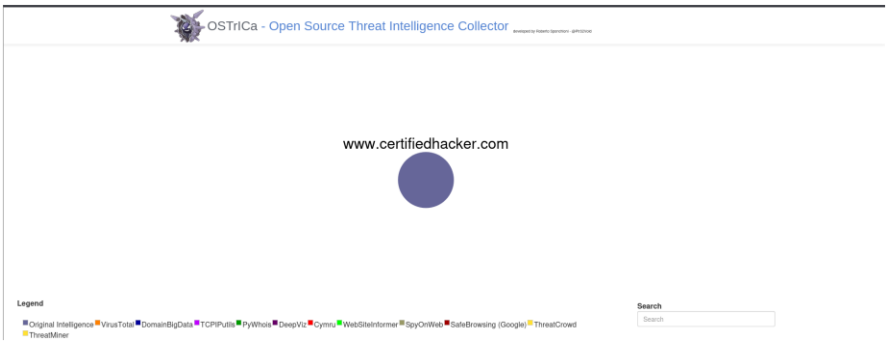
Step 7: Open Generated HTML in Browser

Navigated to the viz folder and opened the HTML file in the browser.



Observation – Graph Display

The intelligence graph loaded in the browser and displayed the domain www.certifiedhacker.com as the central node. However, several plugins (DeepViz, PyWhois, CymruWhois) failed to load due to missing Python modules, and ThreatCrowd encountered an SSL error during the run — this resulted in limited data being collected. Due to the reduced data, the graph rendered as a single consolidated node rather than the expected multi-node relationship map.



Open Source Intelligence Gathering Using OSRFramework

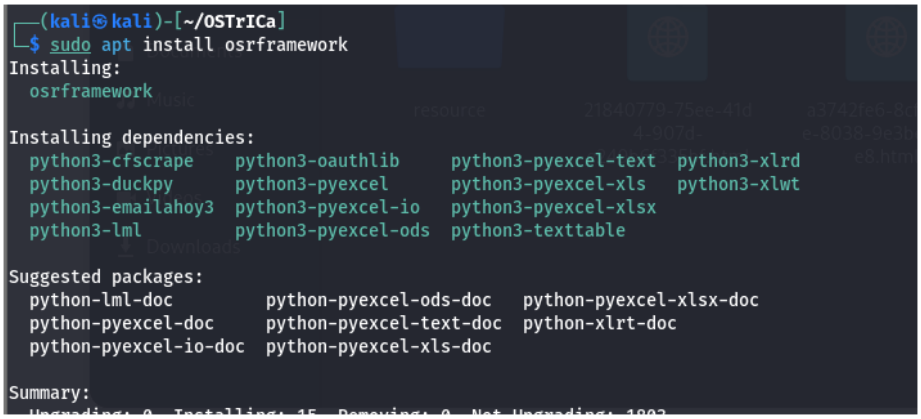
OSRFramework is a Python-based OSINT framework for locating and analyzing user profiles across social media platforms. It includes tools such as usufy.py (profile lookup) and searchfy.py (page/subscriber search).

Step 1: Install OSRFramework

OSRFramework was installed using:

```
sudo apt install osrframework
```

Installation completed successfully with 15 packages installed including Python3 dependencies (pyexcel, oauthlib, texttable, etc.).



Step 2: Check Available Platforms

To identify the correct platform names supported in this version, the following command was run:

```
usufy.py --show_tags
```

The output listed all available platforms grouped by category (e.g., activism, audio, blog, social, etc.). This step was needed because the newer version of OSRFramework no longer accepts 'twitter' or 'facebook' as platform names.





Step 3: Search User Profile Using usufy.py

The following command was used to search for the username 'analyst' across VK, GitHub, and Instagram:

```
usufy.py -n analyst -p vk github instagram
```

```
┌──(root㉿kali)-[/home/kali/OSTrICa]
└─# usufy.py -n analyst -p vk github instagram
```

```
-- In 'domainfy', you may use several words with '-n'. E. g.: '-n felix yaiza'. --

      Usufy | Copyright (C) Yaiza Rubio & Félix Brezo (i3visio) 2014-2021

This program comes with ABSOLUTELY NO WARRANTY. This is free software, and you
are welcome to redistribute it under certain conditions. For additional info,
visit <https://www.gnu.org/licenses/agpl-3.0.txt>.


2026-03-01 12:00:05.806937      Starting search in 3 platform(s)... Relax!

        Press <Ctrl + C> to stop...


2026-03-01 12:00:11.434922      Results obtained (2):

Objects recovered (2026-3-1_12h0m).:
+------------------------------+------------------+----------------------+
|       com.i3visio.URI        | com.i3visio.Alias | com.i3visio.Platform |
+==============================+==================+======================+
| https://github.com/analyst   | analyst          | Github               |
+------------------------------+------------------+----------------------+
| http://instagram.com/analyst | analyst          | Instagram            |
+------------------------------+------------------+----------------------+

2026-03-01 12:00:11.543549      You can find all the information here:
        ./profiles.csv

2026-03-01 12:00:11.543671      Finishing execution...

Total time consumed:    0:00:05.736734
Average seconds/query:  1.9122446666666668 seconds


Did something go wrong? Is a platform reporting false positives? Do you need to
integrate a new one and you don't know how to start? Then, you can always place
an issue in the Github project:
    https://github.com/i3visio/osrframework/issues
Note that otherwise, we won't know about it!
```

Step 4: Search Page Subscribers Using searchfy.py

searchfy.py was used to find all users subscribed to a page named 'certifiedhacker' across social platforms:

```
┌──(root㉿kali)-[/home/kali/OSTrICa]
└─# searchfy.py -q certifiedhacker
```

The tool searched GitHub, Instagram, and KeyServerUbuntu modules. The query returned 'No data found' — this is expected as the search targets public subscriber lists which vary by platform availability and API access.



Conclusion

This lab demonstrated the use of two open-source intelligence (OSINT) tools for data collection and processing:

1. OSTrICa successfully initiated multi-plugin intelligence extraction against www.certifiedhacker.com. Some plugins failed due to missing dependencies and SSL errors, limiting the graph output. The graph was successfully generated and opened in the browser.
2. OSRFramework's usufy.py successfully located the 'analyst' user profile on GitHub and Instagram. searchfy.py was also executed against 'certifiedhacker' as the target page.